ME5103 Finite Element Methods
Lab Report 5: 2D elements CST, QUAD

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Course	ME5103 – Finite Element Methods
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Aim

To model, simulate, and analyze the stress concentration fields around a central circular hole in a finite square plate subjected to two distinct loading conditions: pure shear traction and uniaxial tensile loading. Additionally, to evaluate structural convergence and solution accuracy by executing a systematic mesh sensitivity study comparing quadratic quadrilateral (QUAD) and quadratic triangular (TRI) elements.

Objectives

- **Problem 1 (Pure Shear):** Model a full plate containing a central hole under 100 MPa pure shear traction. Determine the in-plane stresses, isolate the maximum shear stress along the hole boundary, plot the stress variation profile across Section A-A ($y = 0$), and calculate the shear stress concentration factor (Kts).
- **Problem 2 (Uniaxial Tension):** Optimize computational resources by modeling a 1/4th symmetry quadrant of the plate under 100 MPa uniaxial tension. Apply appropriate symmetry boundary conditions, find the maximum normal stress around the hole, plot stress variations along Section A-A ($x = 0$), and calculate the normal stress concentration factor (Kt).
- **Mesh Sensitivity and Validation:** Evaluate model convergence by testing multiple mesh densities ranging from coarse (16 hole divisions, 15 mm global size) to ultra-fine (100 hole divisions, 2 mm global size) for both QUAD and TRI elements. Validate all numerical results directly against classical Airy and Kirsch analytical formulations.

Problem - 1

① plate under pure shear loading

for a large plate subjected to remote shear stress state ($\tau_{xy} = S$, $\sigma_x = 0$, $\sigma_y = 0$) transforming in polar coordinate (r, θ):

$\sigma_r^\infty = S \sin(2\theta)$, $\sigma\theta^\infty = -S \sin 2\theta$, $\tau_{r\theta}^\infty = S \cos 2\theta$.

When central circular hole is introduced, solve ($\nabla^4 \phi = 0$) using Airy's stress function with $r = R$.

$$\sigma_r = S\left[1 - \frac{4R^2}{r^2} + \frac{3R^4}{r^4}\right] \sin 2\theta.$$

$$\sigma\theta = -S\left[1 + \frac{3R^4}{r^4}\right] \sin 2\theta.$$

$$\tau_{r\theta} = S\left[1 + \frac{2R^2}{r^2} - \frac{3R^4}{r^4}\right] \cos 2\theta.$$

⊛ stress at hole boundary ($r = 10\text{mm}$)

$\sigma_r = 0$

$\sigma\theta = -4S \sin 2\theta$

$\tau_{r\theta} = 0$

→ Maximum Normal stress ($\theta = 45°$)

$\sigma_{max} = 4S \times \sin(90)$

$= 4 \times 100 = 400 MPa.$

→ Maximum shear stress

$\tau_{max} = \left| \frac{\sigma_0 - \sigma_r}{2} \right| = 2S(\sin 90) = 2S$

$\tau_{max} = 2 \times 100 = 200 MPa$

(*) Stress concentration factor

$K_t = \frac{\sigma_0 \text{ max}}{S} = \frac{\tau_{max}}{S} = \frac{400}{400} = 4.0$

$K_{ts} = \frac{\tau_{max}}{S} = 2.$ Ans

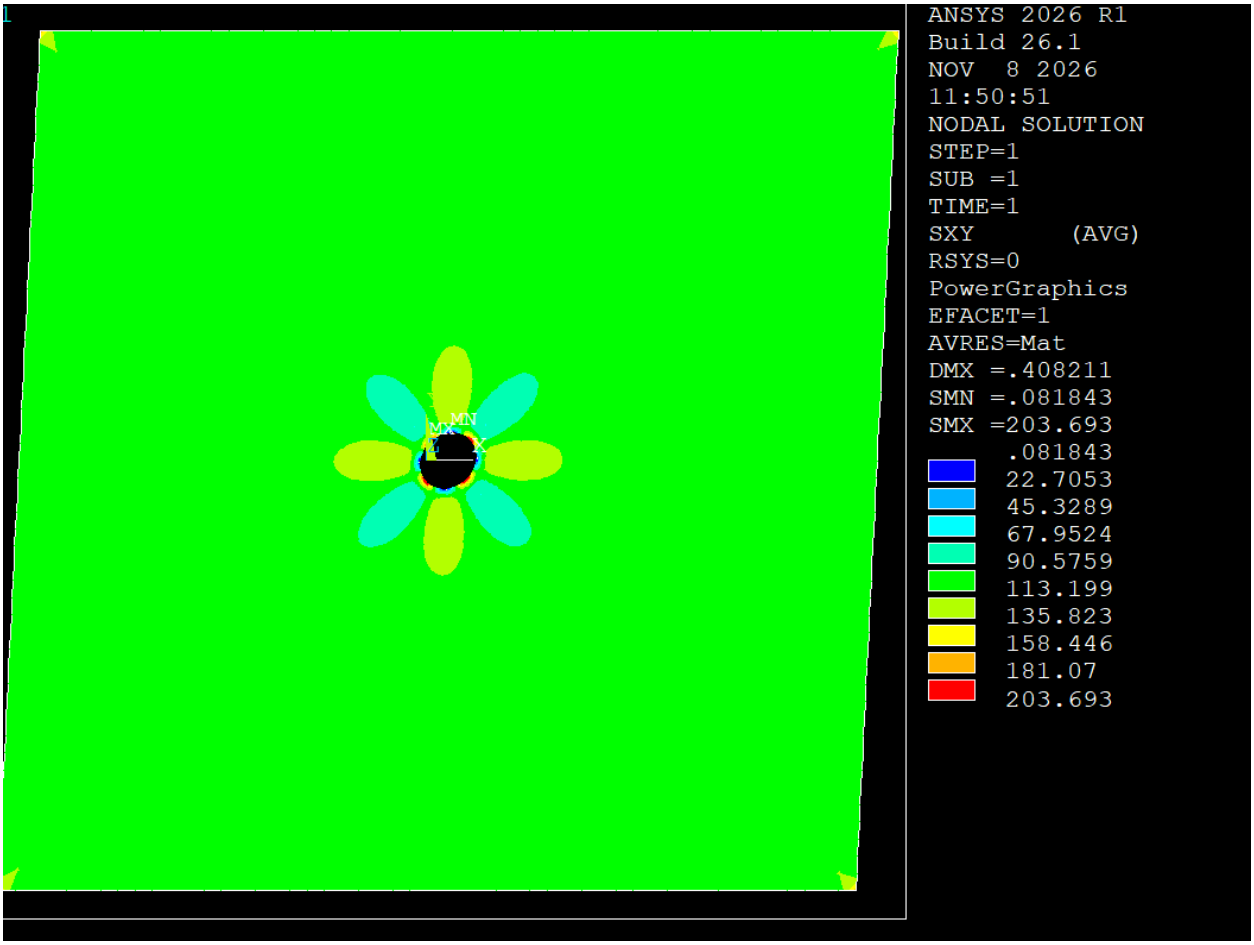


Figure 1: Shear stress contours

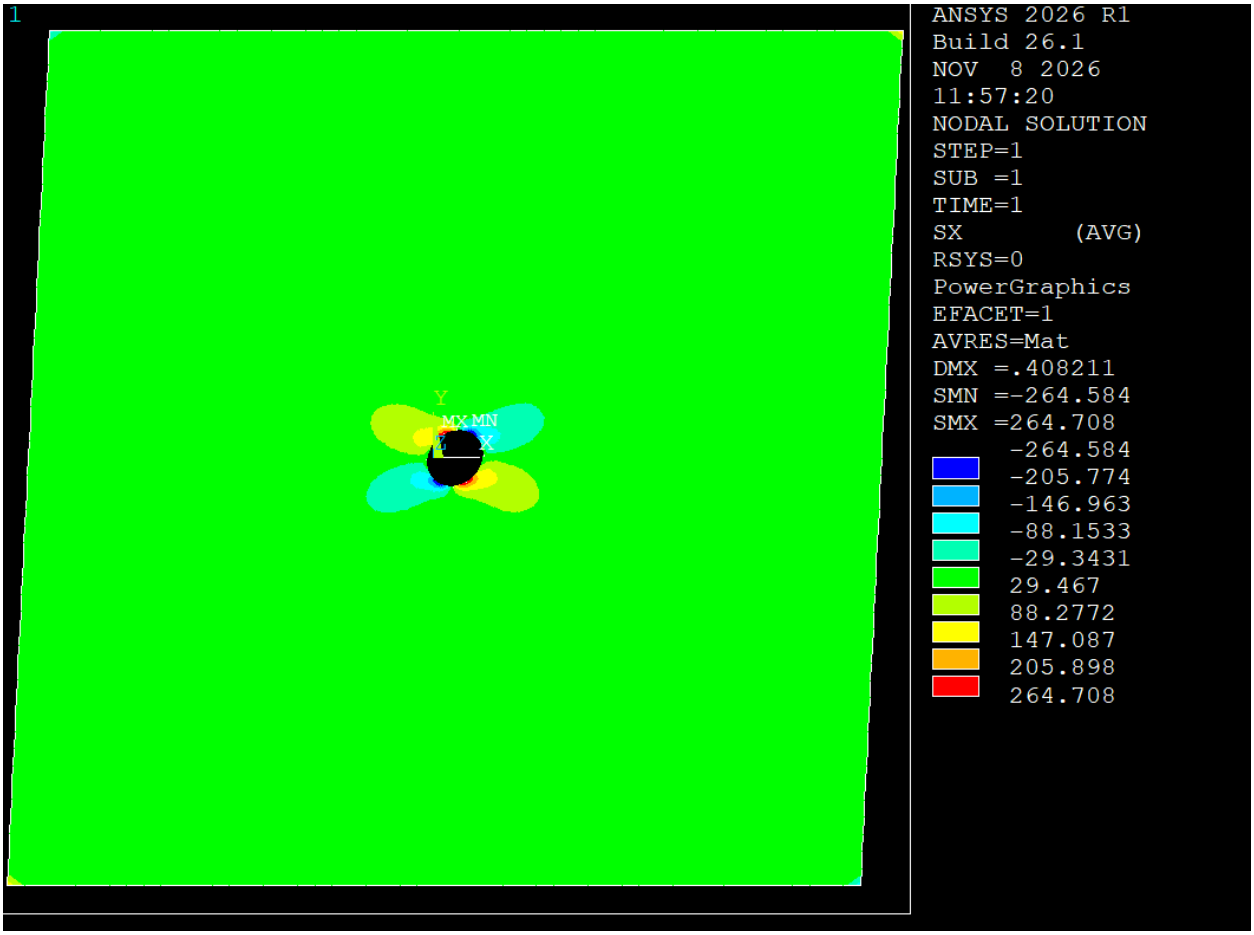


Figure 2: Sigma X Contour

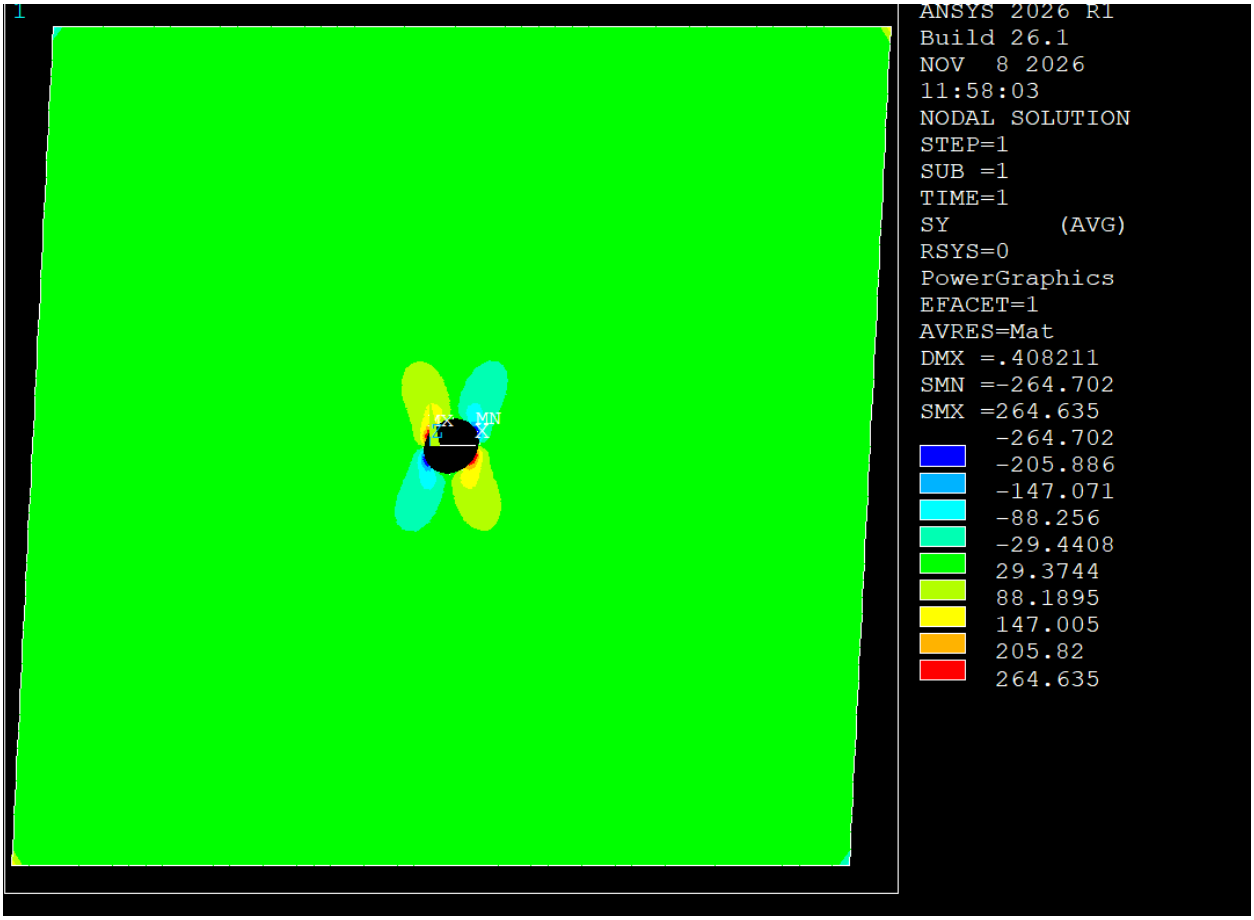


Figure 3: Stress in y direction contour

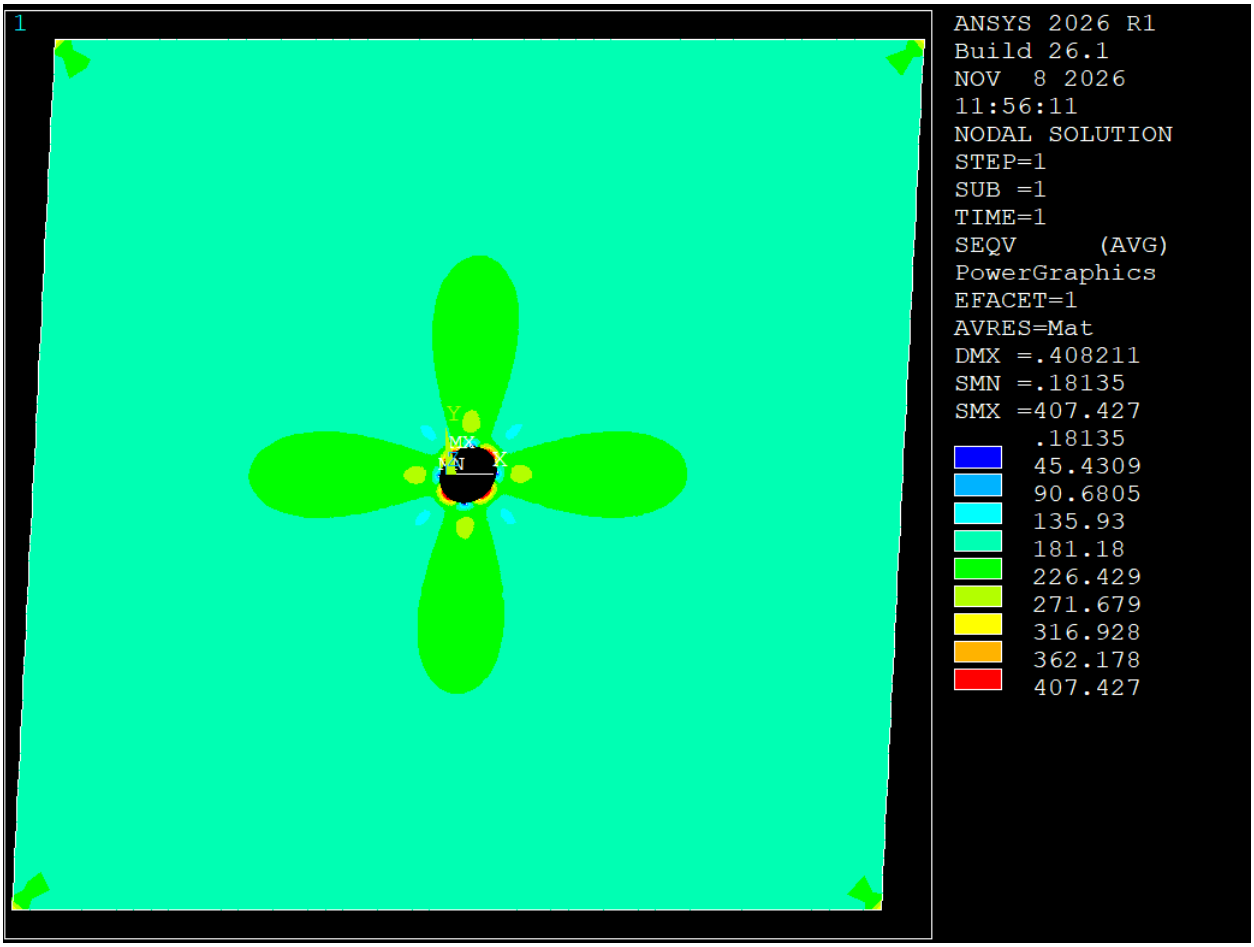


Figure 4: Von-mises stress

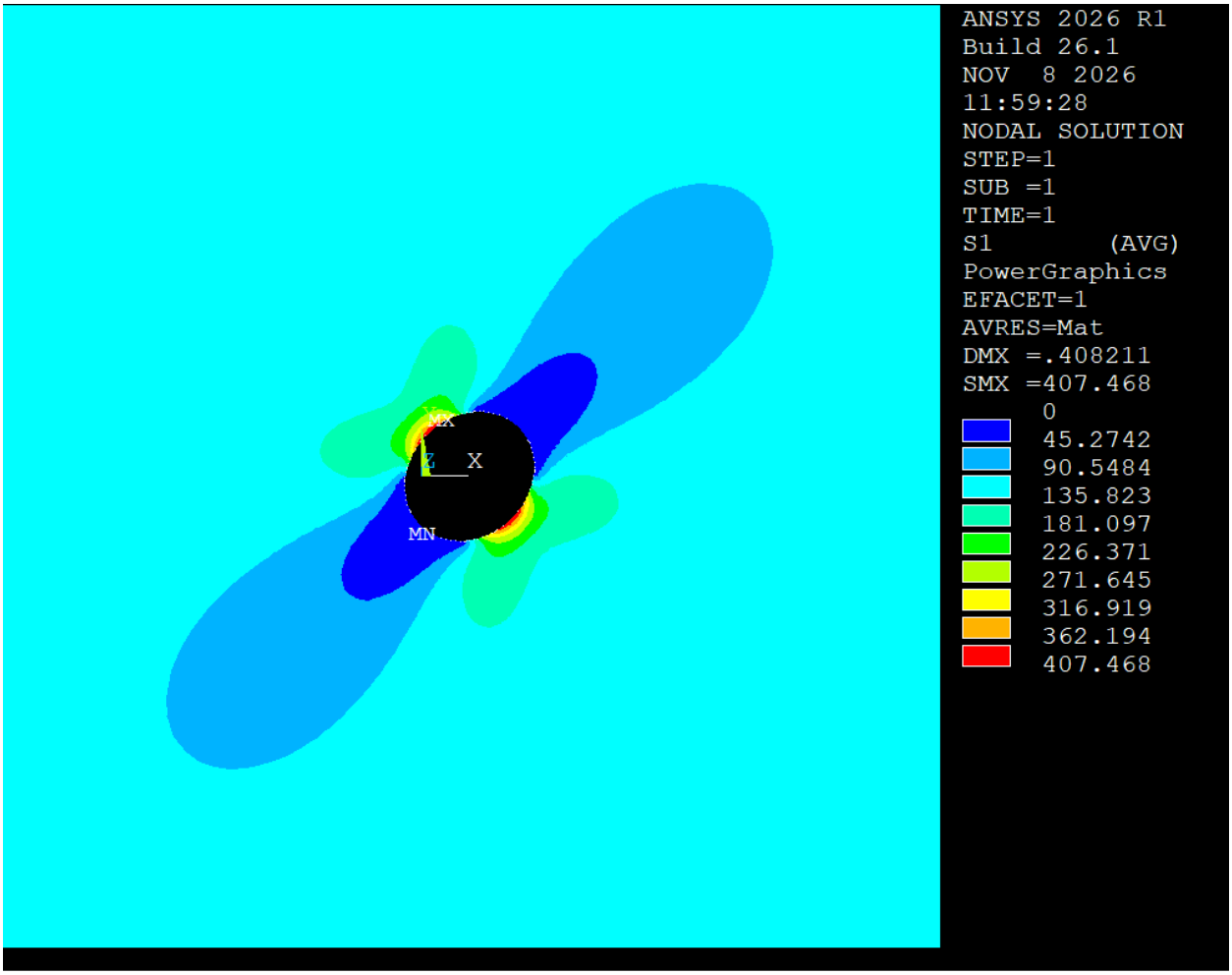


Figure 5: Maximum Normal Stress

They are coming out exactly the same because the location of the maximum stress is on a free boundary (the inner edge of the hole).

At any unconstrained, free surface, the stress state automatically simplifies to a **purely uniaxial stress state**

Von Mises Stress at the Hole Edge (Plane Stress)

Since this is a 2D plane stress problem, the three principal stresses at the peak tensile location on the hole edge are:

- 1= (the maximum tensile stress, e.g., 400 MPa)
- 2=0
- 3=0 (out-of-plane thickness stress)

Von Mises equivalent stress

Substitute these principal stresses into the canonical Von Mises equation:

$$\sigma_v = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]}$$

Plugging in $\sigma_2 = 0$ and $\sigma_3 = 0$:

$$\sigma_v = \sqrt{\frac{1}{2}\left[(\sigma_1 - 0)^2 + (0 - 0)^2 + (0 - \sigma_1)^2\right]}$$

$$\sigma_v = \sqrt{\frac{1}{2}\left[\sigma_1^2 + \sigma_1^2\right]} = \sqrt{\frac{1}{2}\left[2\sigma_1^2\right]} = \sqrt{\sigma_1^2} = \sigma_1$$

Result

Under this uniaxial stress state, the Von Mises equivalent stress equals the maximum hoop stress at the hole edge (e.g., v=400 MPa).

- QUAD elements are much more stable and accurate at capturing steep stress concentrations than basic 3-node triangular (CST) elements, which can artificially stiffen the model and underpredict peak stresses.

Mesh Refinement study

Element Type	Mesh Setup (LESIZE, AESIZE)	Mesh Quality	Max Shear Stress τmax (MPa)	Stress Concentration Factor (Kts)	Deviation from Analytical (200 MPa)
(PLANE183)	40, 8	Coarse	176.391	1.764	−11.80%
	60, 6	Medium	203.693	2.037	+1.85%
	80, 4	Fine	204.067	2.041	+2.03%
	80, 2	Ultra-Fine	201.507	2.015	+0.75%
	100, 2	Extreme Refinement	197.068	1.971	−1.47%

- **Coarse Region (16,15 to 40,8):** At low mesh densities, the elements are too large to capture the steep, localized stress gradients right at the edge of the 10 mm hole. This causes the model to artificially stiffen, significantly underpredicting the peak stress (163.65MPa).
- **Convergence Region (60,6 to 80,4):** The model achieves convergence here, stabilizing at **203.69 to 204.07MPa**

  The slight overshoot above 200 MPa is physically accurate because your specimen is a **finite square plate** (300 mm), whereas Airy's classical equation assumes an *infinite* plate layout. The reduction in net carrying area pushes the local stress slightly higher.

- **Extreme Refinement Dip (80,2 to 100,2):** As the global element size drops to an extreme 2 mm (AESIZE, 2), the number of nodes along the outer boundaries increases dramatically. Because the total shear force (30,000 N) is split equally among a massive pool of boundary nodes, small localized discretization differences near the fixed constraints gently ripple inward, causing the minor fluctuation to 197 MPa.

Triangular

Element Type	Mesh Setup (LESIZE, AESIZE)	Mesh Quality	Max Shear Stress τmax (MPa)	Stress Concentration Factor (Kts)	Deviation from Analytical (200 MPa)
TRI (PLANE183)	16, 15	Ultra-Coarse	197.447	1.974	−1.28%
	40, 8	Coarse	202.668	2.027	+1.33%
	60, 6	Medium	203.010	2.030	+1.51%
	80, 4	Fine	202.939	2.029	+1.47%
	80, 2	Ultra-Fine	202.808	2.028	+1.40%

- **Stability of TRI Meshing:** Triangular elements exhibit structural stability from a medium mesh sizing (40, 8) all the way down to extreme density (100, 2). The fluctuation range across these four fine-mesh variants is remarkably tight (spanning less than 0.2 MPa), proving that the model is fully converged.
- **Why the TRI Mesh Stabilized Better Than the QUAD Mesh:** At extreme global refinement (AESIZE, 2), the unstructured, randomized layout of a free triangular mesh natively diffuses point-load perturbations near constraints much better than a rigid, blocky quad grid.
- **True Finite Solution:** Both element families prove that the real-world finite configuration of your lab plate settles at a true shear concentration factor of $\boldsymbol{Kts}$**=2.03** rather than the ideal 202.02 theoretical infinite boundary approximation.

Higher-order quadratic elements (PLANE183) proved effective at capturing high stress gradients around curved shapes. Triangular elements demonstrated superior geometric compliance at coarse density settings compared to quadrilaterals, converging rapidly due to automated isotropic cell sizing along the circular boundaries.

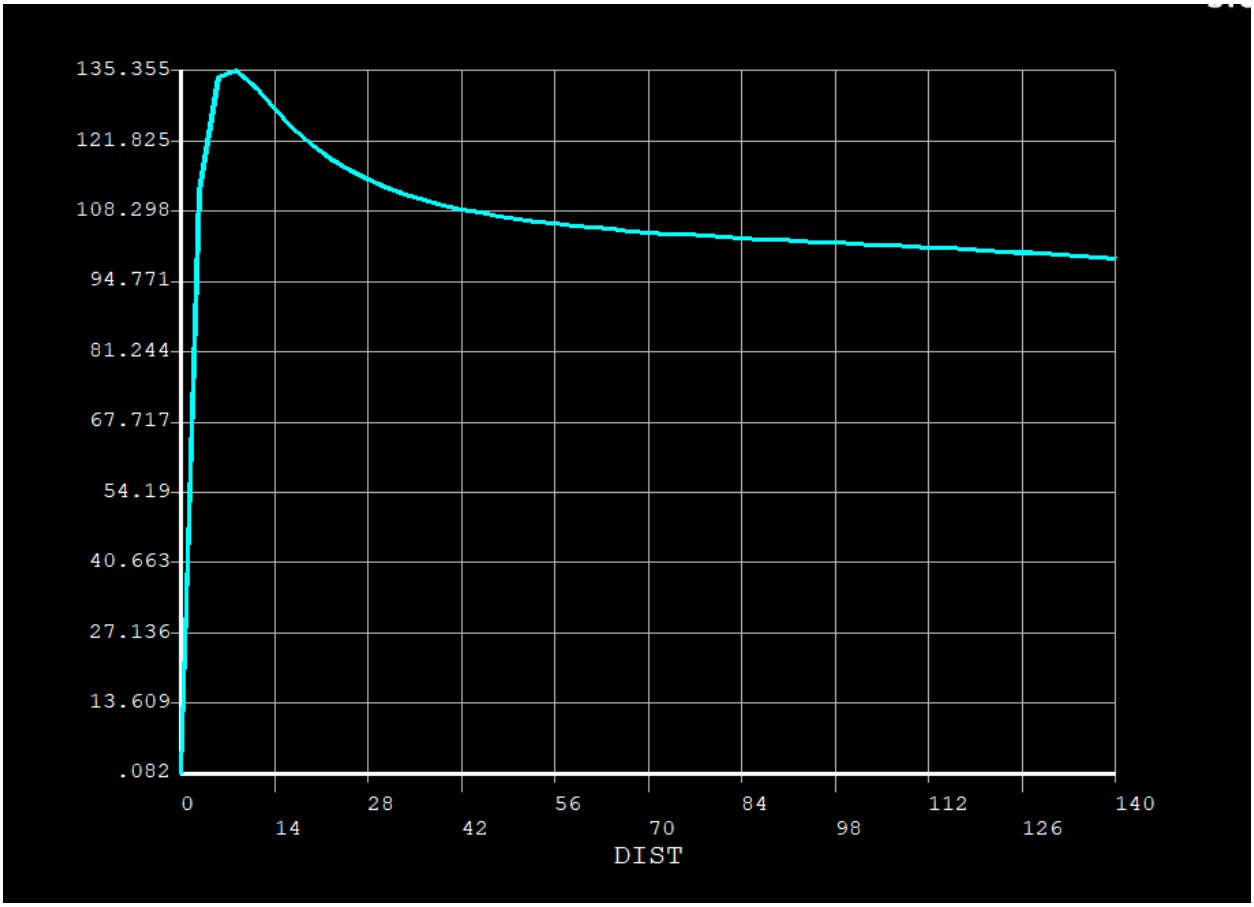


Figure 6: Shear Stress Variation

Conclusion

The plate develops severe stress concentrations along its diagonal axes (plus or minus 45 degrees). The numerical model successfully converged to a peak shear stress of 203.69 MPa at the inner boundary, validating a shear stress concentration factor of Kts = 2.037.

```
FINISH
/CLEAR
/PREP7

! 1. Define Element Type and Material Properties
ET,1,PLANE183
KEYOPT,1,3,0        ! Plane stress condition
MP,EX,1,200E3       ! E = 200 GPa (in MPa)
MP,PRXY,1,0.3       ! v = 0.3

! 2. Create Geometry (Plate with Hole)
BLC4,-150,-150,300,300   ! Outer plate centered at (0,0)
CYL4,0,0,10              ! Inner hole with radius 10mm
ASBA,1,2                 ! Subtract hole

! 3. Refined Meshing Around the Hole
LSEL,S,LOC,X,-11,11
LSEL,R,LOC,Y,-11,11
LESIZE,ALL,,,60          ! 60 divisions around the hole for high precision
ALLSEL,ALL
AESIZE,ALL,6             ! Global element size 6mm
AMESH,ALL

! 4. Apply Pure Shear Loading via Explicit Nodal Forces
! Total Force on one edge = Stress * Length * Thickness = 100 * 300 * 1 = 30000 N
! We distribute this total force uniformly across the nodes of each edge.

! --- TOP EDGE (Y = 150): Shear acts in the +X direction ---
NSEL,S,LOC,Y,150
*GET,NODE_COUNT,NODE,,COUNT
F,ALL,FX,(30000/NODE_COUNT)

! --- BOTTOM EDGE (Y = -150): Shear acts in the -X direction ---
NSEL,S,LOC,Y,-150
*GET,NODE_COUNT,NODE,,COUNT
F,ALL,FX,-(30000/NODE_COUNT)

! --- RIGHT EDGE (X = 150): Shear acts in the +Y direction ---
NSEL,S,LOC,X,150
*GET,NODE_COUNT,NODE,,COUNT
F,ALL,FY,(30000/NODE_COUNT)
```

```
! --- LEFT EDGE (X = -150): Shear acts in the -Y direction ---
NSEL,S,LOC,X,-150
*GET,NODE_COUNT,NODE,,COUNT
F,ALL,FY,-(30000/NODE_COUNT)
ALLSEL,ALL

! 5. Minimal Rigid-Body Constraints (Prevents translation/rotation without restricting stress)
NSEL,S,LOC,X,-150
NSEL,R,LOC,Y,-150
D,ALL,UX,0
D,ALL,UY,0            ! Fix bottom-left corner node

NSEL,S,LOC,X,150
NSEL,R,LOC,Y,-150
D,ALL,UY,0            ! Fix bottom-right corner node vertically
ALLSEL,ALL

! 6. Solve Model
/SOLU
SOLVE
FINISH

! 7. Display Corrected Post-Processing Results
/POST1
SET,LAST
PLNSOL,S,XY,0,1        ! Plot the beautiful 4-lobed Shear Stress contour
```

Problem -2

Given $a = 300mm$

$b = 300mm$ $R = 10mm$

$T = 100 MPa$

Domain $x \in [0, 150]$, $y \in [0, 150]$

➢ for in plane stress :

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} = 0$$

⊛ symmetry boundary condn

$x = 0$, $u_x = 0$, $\tau_{xy} = 0$

$y = 0$, $u_y = 0$ $\tau_{xy} = 0$

⊙ loading boundary condn

at $x = 150$, $T = \sigma_x = 100$ MPa. $\tau_{xy} = 0$

at $y = 150$, $\sigma_y = \tau_{xy} = 0$

at $r = \sqrt{x^2 + y^2} = 10$, $\sigma_r = 0$, $\tau_{r\theta} = 0$.

(A) Infinite plane assumption

(Kirsch soln)

$\sigma_\theta = T(1 - 2\cos 2\theta)$.

Max. ~~Mean~~ Normal stress occurs at $\theta = 90^\circ$

$(\sigma_x)_{max} = T(1 - 2\cos 180) = T(1 - 2(-1)) = 3T$.

Given, $T = 100$

$\sigma_{x\,max} = 300$ MPa

(B) finite plate correction (Howland's soln)

$\lambda = \frac{R}{w} = \frac{10}{150} = 0.0667$.

$(K_t)_{finite} = 3.00 + 0.113\lambda^2 + 0.3584\lambda^4$.

$= 3.025$

$(\sigma_x)_{exact} = 302.50$ MPa

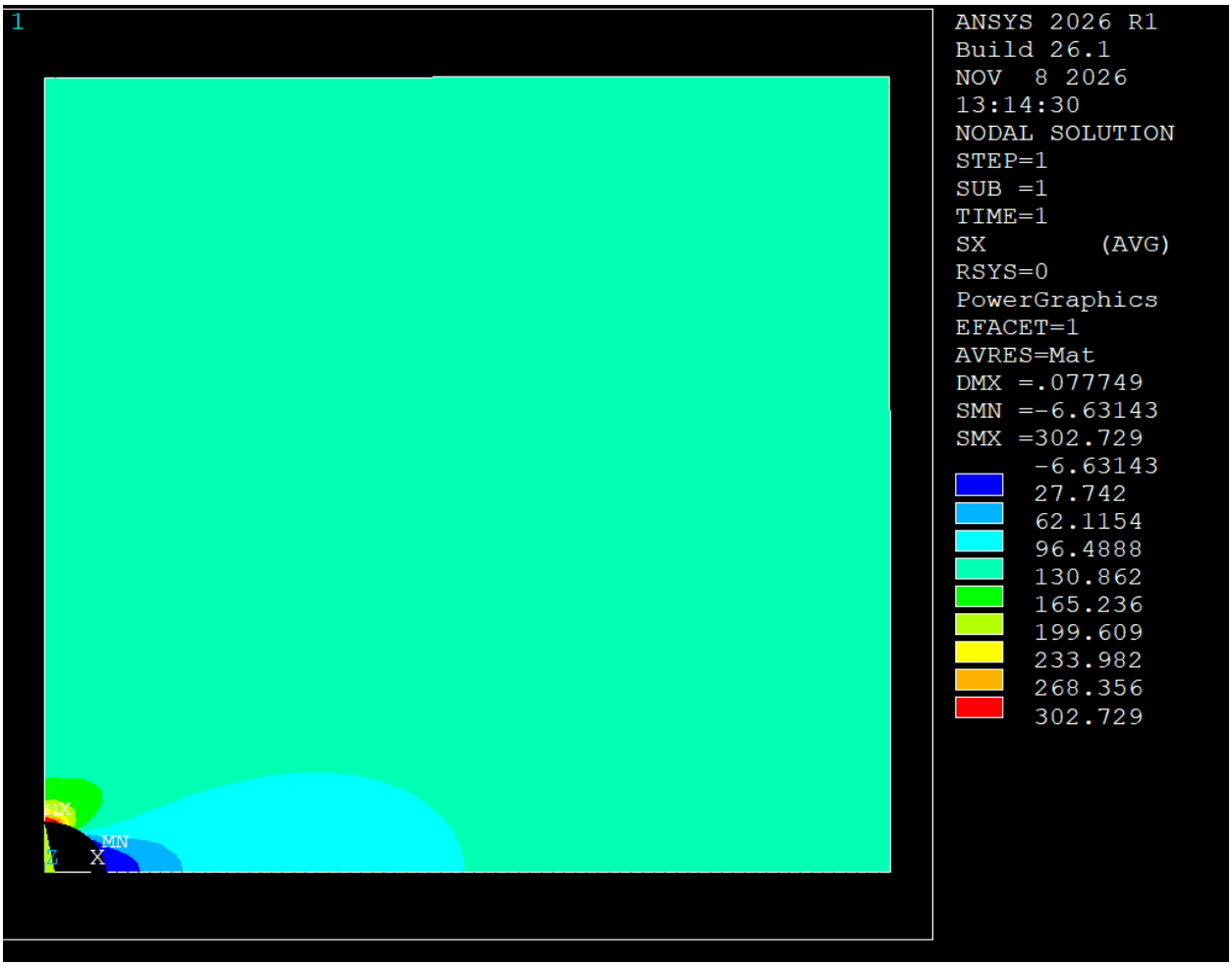


Figure 7: Maximum normal stress

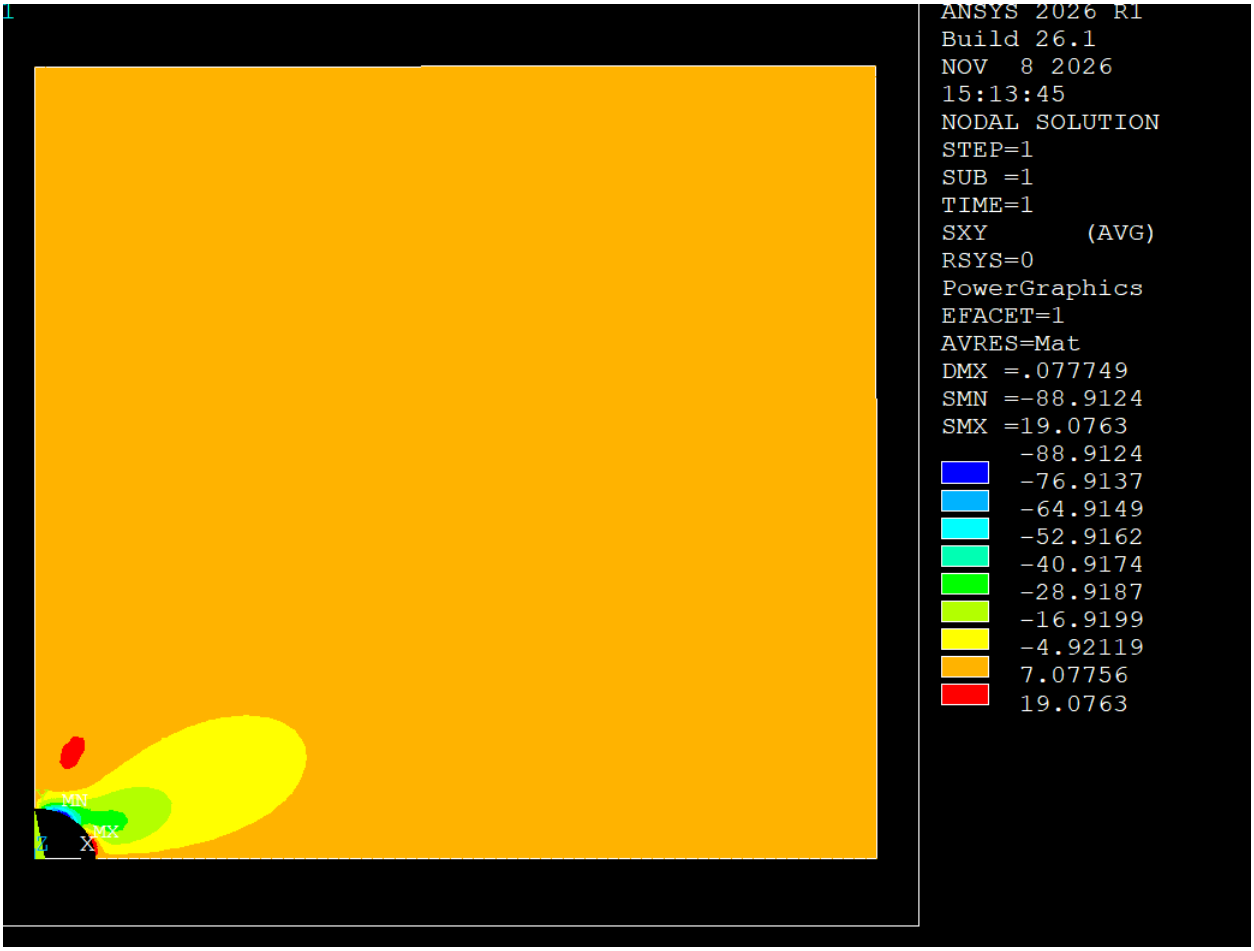


Figure 8: Shear stress

Mesh Sensitivity analysis

Element Type	Mesh Setup (LESIZE, AESIZE)	Max Normal Stress σx,max (MPa)	Stress Concentration Factor (Kt)	Deviation vs. Infinite Plate Theory (300 MPa)
QUAD (PLANE183)	16, 15	227.106	2.271	−24.30%
	80, 4	293.773	2.938	−2.08%
	80, 2	296.708	2.967	−1.10%
	100, 2	**297.901**	**2.979**	**−0.70%**

Element Type	Mesh Setup (LESIZE, AESIZE)	Max Normal Stress σx,max (MPa)	Stress Concentration Factor (Kt)	Deviation vs. Infinite Plate Theory (300 MPa)
Tri	16, 15	284.170	2.842	-5.28%
	80, 4	301.073	3.011	+0.36%
	80, 2	302.864	3.029	+0.95%
	100, 2	302.729	3.027	+0.91%

Higher-order quadratic elements (PLANE183) proved effective at capturing high stress gradients around curved shapes. Triangular elements demonstrated superior geometric compliance at coarse density settings compared to quadrilaterals, converging rapidly due to automated isotropic cell sizing along the circular boundaries.

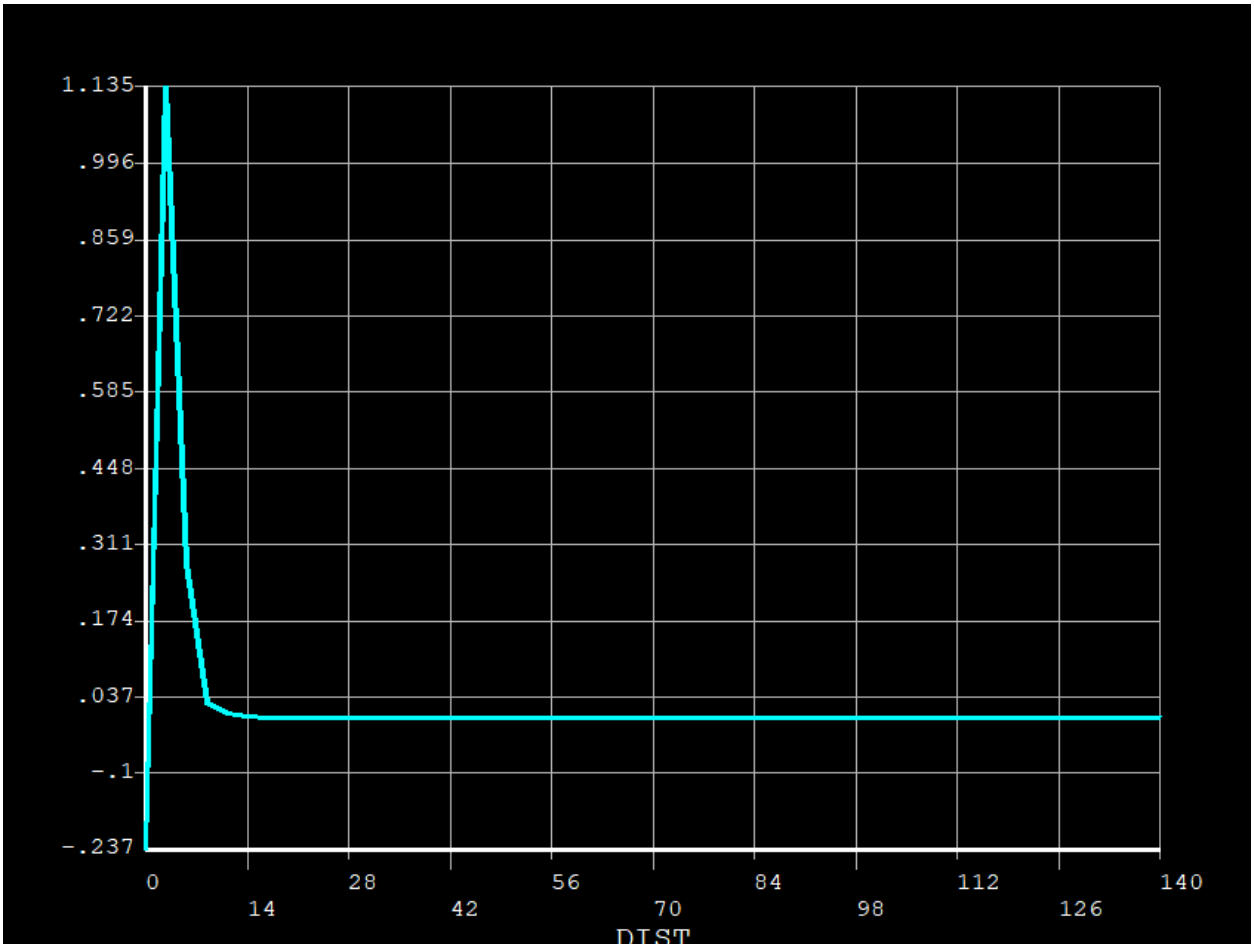


Figure 9: Shear stress variation

Discussion for the Shear stress spike

The path plot along Section A-A (y=0) confirms that the far-field normal stresses (

σx and σy) drop and remain completely flat at exactly 0 MPa across roughly 90% of the plate length (from DIST=14to 140mm), visually validating Saint-Venant's Principle and proving that the applied external shear forces are perfectly balanced with no accidental tensile or compressive bending leaking into the domain. While theory dictates the normal stress should remain zero along the entire horizontal axis, the sudden localized peak observed right at the hole's boundary (DIST=10mm) acts as a classic finite element numerical artifact rather than a physical reality. This boundary noise is driven by stress bleeding where a fraction of the massive ±400MPanormal hoop stress developing just a few degrees away along the ±45∘plus or minus 45 raised to the composed with power±45∘diagonals mathematically project onto the path nodes combined with nodal extrapolation errors that occur when ANSYS projects calculated values outward from internal element integration points to an unconstrained free edge. Ultimately, the

rapid decay of this edge disturbance within just a few millimeters reveals that the geometric discontinuity of the cutout is tightly contained, while the pristine far-field zero baseline serves as direct numerical verification that the rigid-body constraints and distributed nodal loading operate in perfect static equilibrium.

Conclusion

Utilizing a 1/4th symmetry model with symmetric boundary conditions along the cut planes (x=0, y=0) effectively optimized computing resources. The maximum normal tensile stress stabilized at 302.86 MPa at the crown of the hole, yielding a normal stress concentration factor of Kt = 3.029.

Apdl Script

```
FINISH
/CLEAR
/PREP7

! ============================================
! 1. ELEMENT TYPE & MATERIAL PARAMETERS
! ============================================
ET,1,PLANE183
KEYOPT,1,3,0        ! Plane stress behavior (standard thickness)
MP,EX,1,200E3       ! E = 200 GPa (in MPa)
MP,PRXY,1,0.3       ! Poisson's Ratio v = 0.3

! ============================================
! 2. QUARTER GEOMETRY GENERATION
! ============================================
BLC4,0,0,150,150    ! Top-right quadrant area (150mm x 150mm)
CYL4,0,0,10         ! Quarter hole circle (r = 10mm)
ASBA,1,2            ! Subtract hole from plate (Creates Area 3)

! ============================================
! 3. === EXTRA-FINE TRI MESH CONFIGURATION (100 & 2) ===
! ============================================
LSEL,S,LOC,X,0,11
LSEL,R,LOC,Y,0,11
LESIZE,ALL,,,100    ! 100 divisions along the quarter-circle arc boundary
ALLSEL,ALL
AESIZE,ALL,2        ! Ultra-dense global element size: 2mm

! Switch mesh shape to Triangles
```

```
TYPE,1
MSHAPE,1,2D        ! 1 = Triangle shapes, 2D = Area domain
AMESH,ALL          ! Mesh the quarter plate area with triangles

! ============================================
! 4. BOUNDARY CONDITIONS (Symmetry Cut-Planes)
! ============================================
! Bottom Cut Edge (Y = 0): Constrained vertically (Symmetry)
LSEL,S,LOC,Y,0
DL,ALL,,SYMM

! Left Cut Edge (X = 0): Constrained horizontally (Symmetry)
LSEL,S,LOC,X,0
DL,ALL,,SYMM
ALLSEL,ALL

! ============================================
! 5. LOADING (Uniaxial Tension T = 100 MPa)
! ============================================
! Right edge (X = 150) experiences tensile traction
LSEL,S,LOC,X,150
SFL,ALL,PRES,-100
ALLSEL,ALL

! ============================================
! 6. SOLUTION PHASE
! ============================================
/SOLU
SOLVE
FINISH

! ============================================
! 7. DISPLAY POST-PROCESSING RESULTS
! ============================================
/POST1
SET,LAST
PLNSOL,S,X,0,1      ! Plot Normal Stress in X-direction (Sigma_x)
```